National Taiwan University
Department of Economics
ECON 4037: Game Theory
Homework 4: Weeks 9–10
Due 2:20 PM on Wednesday, November 18
You may submit online or hand in a handwritten copy.

Show your work. You may discuss these problems with classmates or use generative AI tools, but you must write your own answers and be able to explain and defend everything you submit (see the syllabus).

This assignment covers Nash-equilibrium applications, Chapters 14–15, and selected applications from Chapters 16–17. In Problems 2–5, find only pure-strategy equilibria.

1. **Protecting three sites.** An attacker chooses one of three sites, A, B, and C, while a defender simultaneously chooses one site to protect. An attack on an unprotected site causes damage of 4 at A, 2 at B, or 1 at C. An attack on a protected site causes no damage. The attacker's payoff equals the damage, and the defender's payoff is its negative. Thus, the attacker maximizes expected damage while the defender minimizes it.

   (a) Write the normal-form payoff matrix and explain why there is no pure-strategy Nash equilibrium.

   (b) Find a mixed-strategy Nash equilibrium. You may begin by looking for an equilibrium in which both players use only A and B, but you must check deviations to C.

   (c) Calculate expected damage in your equilibrium. Explain why the attacker can attack the less valuable site more often than the more valuable site.

2. **From a tree to subgames (Watson Ch. 15, Exercise 2(a)).** Player 1 first chooses W or Z, which player 2 observes. After W, player 2 chooses A or B. Player 1 then chooses X or Y without observing whether player 2 chose A or B. The payoffs, in the order (player 1, player 2), are $(3, 0)$ after (W, A, X), $(8, 5)$ after (W, A, Y), $(4, 6)$ after (W, B, X), and $(2, 1)$ after (W, B, Y). If player 1 instead chooses Z, player 2 chooses C or D and the game ends. Choosing C gives payoffs $(6, 4)$, while choosing D gives payoffs $(3, 2)$.

   (a) Draw the extensive form, including the information set. Does the game have perfect information? Does it have perfect recall? Explain.

   (b) List each player's complete pure strategies and construct the normal-form matrix. Use the notation WX, for example, for player 1's strategy of choosing W initially and X at the later information set.

   (c) Find all pure-strategy Nash equilibria.

   (d) Identify every proper subgame and find all pure-strategy subgame perfect equilibria. For each Nash equilibrium that fails this test, identify a continuation that is not a Nash equilibrium of its subgame.

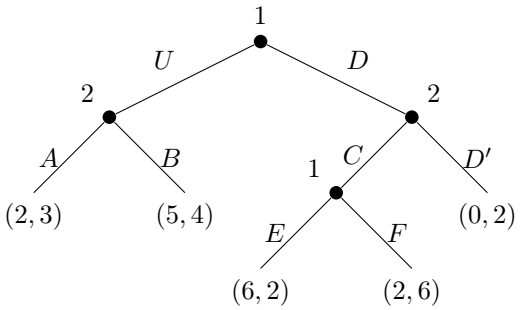
3. **Stopping a growing surplus.** Two players take turns, with player 1 moving at decision nodes 1 and 3 and player 2 moving at nodes 2 and 4. At each node, the mover chooses Stop or Continue. Stopping at nodes 1, 2, 3, and 4 gives payoffs $(2, 1)$, $(1, 4)$, $(5, 3)$, and $(4, 7)$, respectively. Continuing at node 4 ends the game with payoffs $(8, 6)$. All previous actions are observed.

(a) Draw the game tree. Use S_t and C_t to label Stop and Continue at node t.

(b) Find the subgame perfect equilibrium by backward induction. Report complete strategies and the equilibrium outcome.

(c) Find one Nash equilibrium that is not subgame perfect. Verify that neither player can gain by deviating, then identify a continuation that fails sequential rationality. Compare its outcome with the outcome in part (b).

4. **Backward induction with indifference (Watson Ch. 15, Exercise 2(b)).** The tree below describes a game with perfect information. The number at each decision node identifies the player who moves, and payoffs are ordered (player 1, player 2).



(a) List each player's complete pure strategies and identify all proper subgames.

(b) Use backward induction, retaining all optimal actions at ties, to find all pure-strategy subgame perfect equilibria. Report their outcomes and explain why perfect information need not imply a unique subgame perfect equilibrium.

5. **Quantity commitment and entry (adapted from Watson Ch. 16, Exercise 4).** An incumbent chooses a quantity $q_1 \in [0, 12]$. After observing q_1, a potential entrant chooses whether to enter. If it enters, it pays a fixed cost $F > 0$ and chooses $q_2 \in [0, 12]$. Both firms have zero variable costs and sell at the market price $P = 12 - q_1 - q_2$. This price formula applies to all feasible quantity pairs. The incumbent's payoff is Pq_1. The entrant's payoff is $Pq_2 - F$ if it enters and zero if it stays out. When it stays out, $q_2 = 0$. For parts (a)–(c), use the following tie-breaking rule: the entrant stays out whenever entering and staying out give equal payoffs.

(a) Find the entrant's optimal quantity conditional on entry, and its entry decision as a function of q_1 and F.

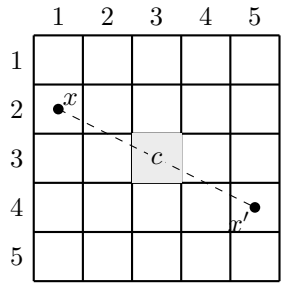
(b) Let $F = 4$. Find a subgame perfect equilibrium consistent with the stated tie-breaking rule. Report complete strategies, equilibrium quantities, price, and both firms' profits. Check the incumbent's choices that accommodate entry as well as those that deter it.

(c) Use the response functions from part (a) to solve the game for $F = 1/4$ and $F = 16$. Report equilibrium quantities, price, and both firms' profits for each fixed cost. Explain how the fixed cost changes the incumbent's incentive to accommodate entry, increase output to deter entry, or produce its monopoly quantity.

(d) Let $F = 4$. Instead of the stay-out rule used above, select a continuation in which the entrant enters whenever it is indifferent. Can a pure-strategy subgame perfect equilibrium use this continuation? Explain what happens to the incumbent's best response.

6. **Mirroring on a board (adapted from Watson Ch. 17, Exercise 7).** Players take turns claiming one empty cell of an $m \times n$ rectangular board, with player 1 moving first. Player 1 uses white chips and player 2 uses black chips. Chips cannot be moved or removed. Play ends when every cell is filled. Each player observes the entire board.

   A cell is a *victory cell* if at least two of its side-sharing neighbors have the same color as that cell; all other cells are *loss cells*. Diagonal neighbors do not count. Each player scores one point for each of their own victory cells and each of the opponent's loss cells. The higher score wins; equal scores result in a draw. Number the rows $1, \ldots, m$ from top to bottom and the columns $1, \ldots, n$ from left to right. A *half-turn* maps (i, j) to $(m+1-i, n+1-j)$. The diagram illustrates the coordinates on a 5×5 board: c is the center, and $x = (2, 1)$ and $x' = (4, 5)$ are a reflected pair. The dots identify cell locations on an otherwise empty board.



   (a) Suppose at least one of m, n is even. Show that player 2 can guarantee a draw by claiming the half-turn image of the cell that player 1 just claimed. Explain why this response is always available and compare final scores.

   (b) Suppose $m, n \geq 3$ are both odd. Show that player 1 can guarantee a win by first taking the center and then using the same response rule. In comparing reflected pairs, treat neighbors of the center separately.

   (c) Now require at least *three* same-color neighbors for a victory cell. On a 3×3 board, does the strategy in part (b) still guarantee a win or a draw? Construct a completed board attainable under this strategy and compute the scores to justify your answer. Explain what your example establishes about this strategy and what it leaves open about other strategies.

7. **Taking the last token.** Two players alternate taking one or two tokens from a pile. A player cannot take more tokens than remain. The player who takes the last token loses. All moves are observed, and each player prefers winning to losing.

   (a) Classify the positions with 1 through 8 tokens as winning or losing for the player whose turn it is, assuming optimal play.

   (b) Give a rule for every positive number of tokens and justify it by backward induction. Specify a winning move whenever one exists.

   (c) Who can guarantee a win with 13 tokens initially? What changes if the initial pile has 14 tokens? Describe the winning player's plan.